

Module 8: Storage and Archiving

At the end of this module, students will be able to:

- **Differentiate** the AWS data storage options and **explain** their purpose and benefits.
- **Create** and **manage** Amazon EBS snapshots.
- **Store, retrieve** and **archive** Amazon S3 objects.
- **Identify** AWS data migration services.

Topics:

- Cloud storage overview
- Amazon EBS
- Instance Store
- Amazon Elastic File System (Amazon EFS)
- Amazon S3
- Amazon S3 Glacier
- AWS Storage Gateway
- AWS data transfer and migration services

Knowledge Check

• Lab:

Lab 5: Managing storage in AWS

• Exercises/Project:

- Hands-on Activity 8: Work with Amazon S3
- Project: Troubleshooting playbook

Troubleshooting knowledge base – Module 8 focus

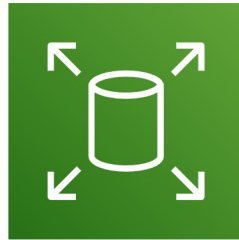
- Have your copy of the troubleshooting knowledge base template open for editing
- **Create new entries** as you encounter troubleshooting situations
- **Categories of focus** for this module:
 - **Storage & Data Management**

Module 8: Storage and Archiving

Cloud storage overview

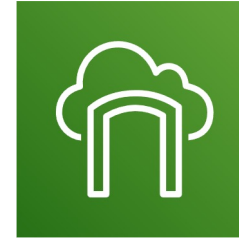
Cloud storage in AWS

Block storage service



Amazon Elastic Block Store (EBS) Instance Store

Hybrid cloud storage service



AWS Storage Gateway

File storage services



Amazon Elastic File System



Amazon FSx for Lustre



Amazon FSx for Windows File Server

Object storage services

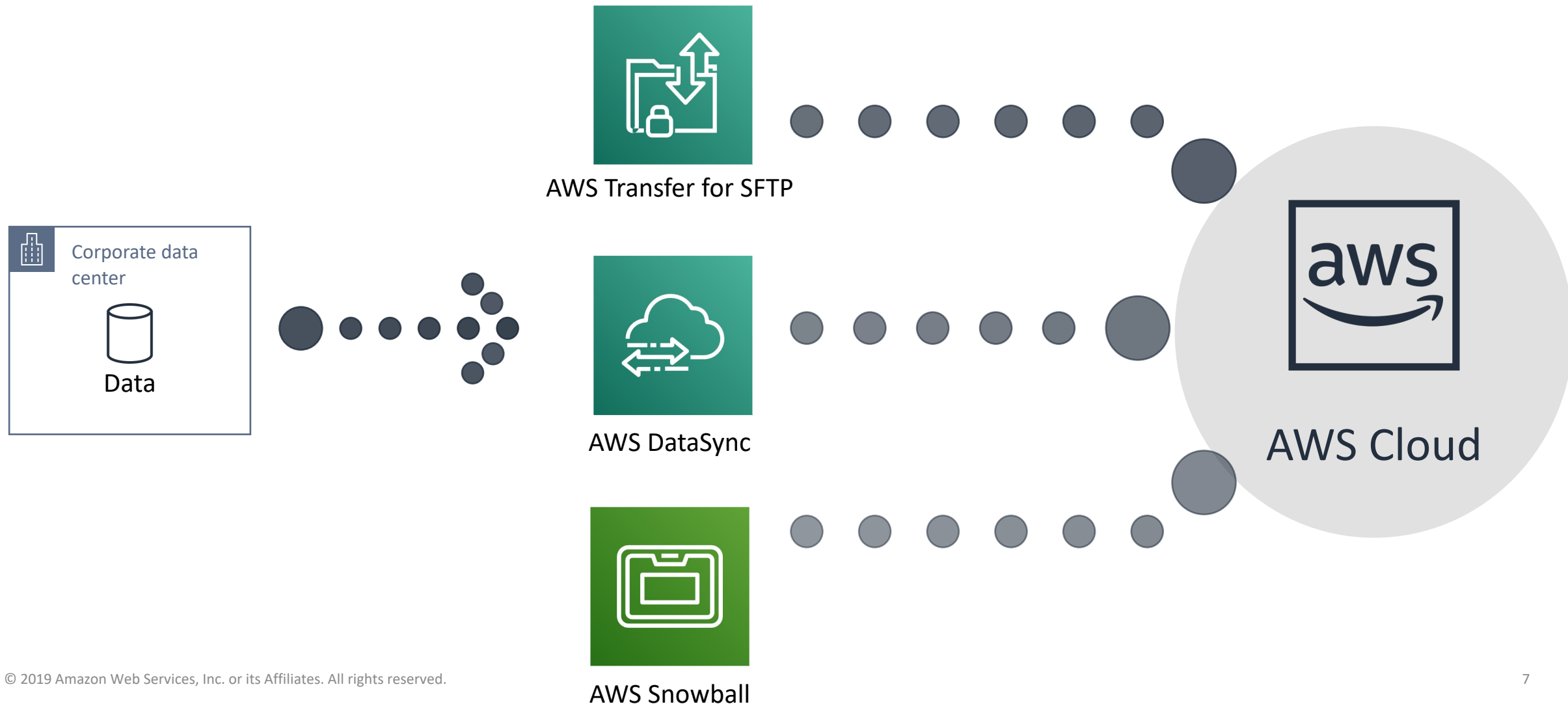


Amazon Simple Storage Service (S3)



Amazon S3 Glacier

Data transfer and migration in AWS



AWS cloud storage scenarios (1 of 2)

If you need:	Consider using:
<i>Persistent block storage</i> for Amazon EC2. Ideal for databases and applications.	Amazon Elastic Block Store (Amazon EBS)
A scalable, durable platform to make data accessible from any Internet location. Ideal for user-generated content, active archive, serverless computing, big data storage or backup and recovery.	Amazon Simple Storage Service (Amazon S3)
Secure, durable, and extremely low-cost cloud storage service for data archiving and long-term backup.	Amazon S3 Glacier

AWS cloud storage scenarios (2 of 2)

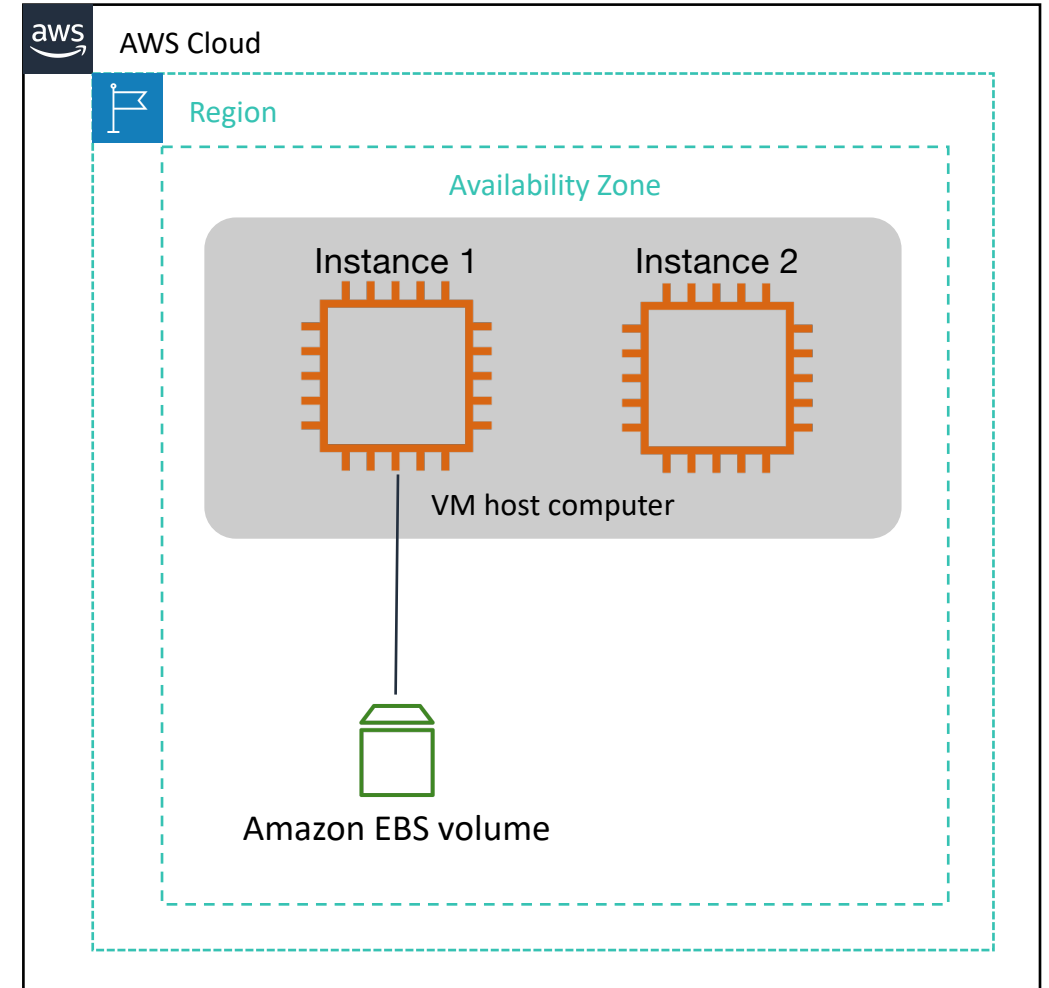
If you need:	Consider using:
A simple, scalable, NFS file system for <u>Linux-based</u> workloads for use with AWS Cloud services and on-premises resources	Amazon Elastic File System (Amazon EFS)
Hybrid storage cloud augmenting your on-premises environment with Amazon cloud storage, for disaster recovery, tiering or migration	AWS Storage Gateway

Module 8: Storage and Archiving

Amazon Elastic Block Store (Amazon EBS)

Amazon EBS volumes are:

- Network-attached
- Persistent block storage
- Boot and data volumes supported



Solid State Drives (SSD)

**Provisioned IOPS
SSD (io1) volumes**

**General Purpose
SSD (gp2) volumes**

Hard Disk Drives (HDD)

**Throughput
Optimized HDD (st1)
volumes**

**Cold HDD
(sc1) volumes**

Create an Amazon EBS volume

- A volume exists in an Availability Zone.
- AWS Command Line Interface (AWS CLI) example:
 - Use the `create-volume` command
 - Use the `--availability-zone` option to specify the Availability Zone

```
$ aws ec2 create-volume --size 80 \  
--availability-zone us-east-1a --volume-type gp2
```

Describe new volume just created

CLI command for describing the volume created:

```
$ aws ec2 describe-volumes --volume-ids vol-0e5e5193f8d093d5f
```

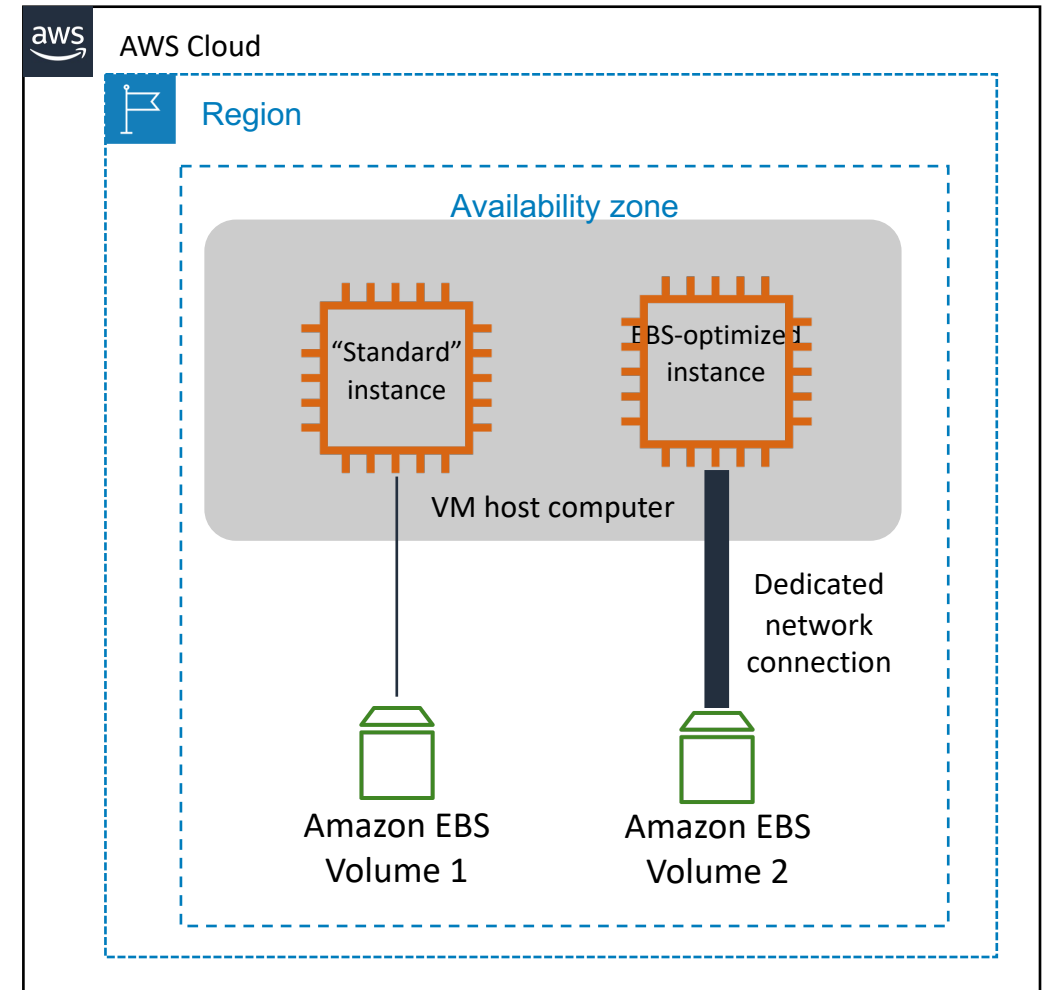
Attach an Amazon EBS volume

- Attach volumes to logical mount points in the operating system
- AWS CLI example:
 - Use the **attach-volume** command.
 - Use the **-- device** option to specify the mount point.

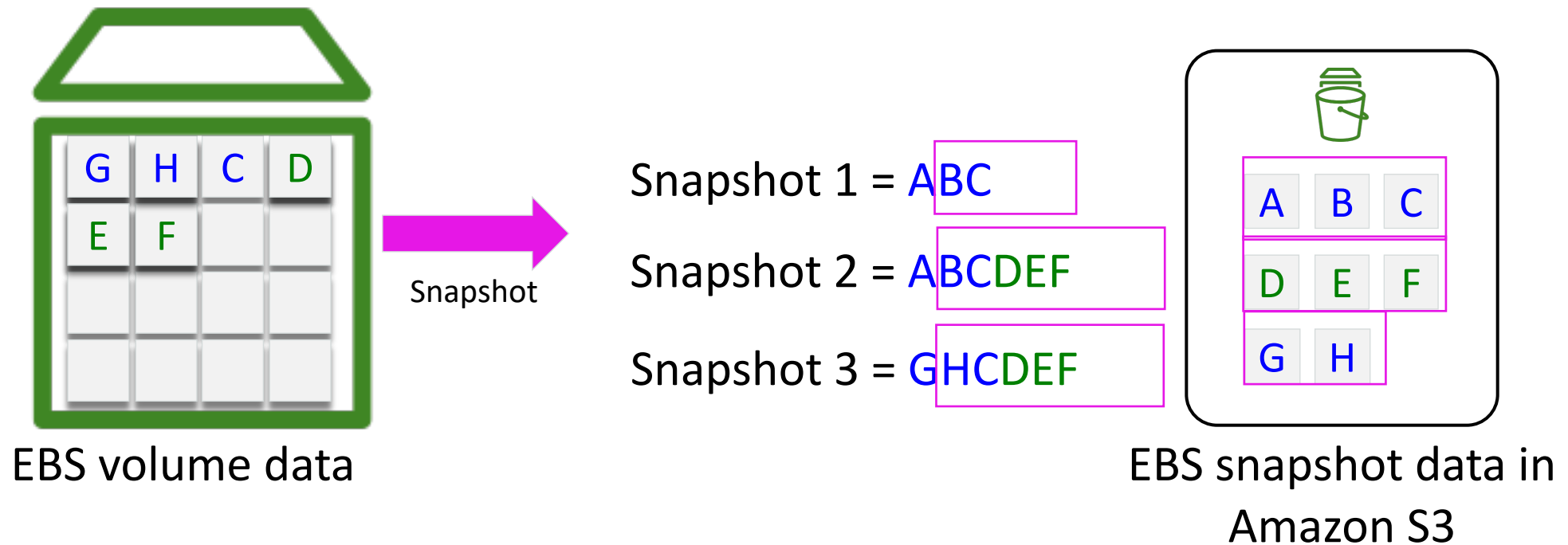
```
$ aws ec2 attach-volume --volume-id vol-12345678 \  
--instance-id i-abc1234 --device /dev/sdf
```

Amazon EBS-optimized instances

- Certain EC2 instance types can be EBS-optimized.
 - Provides a dedicated network connection to EBS volume
 - Increases performance of Amazon EBS I/O
- For EBS-optimized instance types, optimization is enabled by default.
- For other instances types that support it, optimization must be manually enabled.
- Additional performance is achieved if using an Amazon EC2 Nitro System based instance type.



Amazon EBS incremental snapshots



Create a snapshot of an EBS volume

- Use the AWS CLI command **create-snapshot**. The command returns asynchronously.
- Consider stopping the instance or unmounting the volume first:
 - For completeness (otherwise, no subsequent writes are captured.)
 - Strongly recommended for volumes that serve as a root device
 - Critical for database servers and Redundant Array of Independent Disks (RAID) configurations

```
aws ec2 create-snapshot --volume-id vol-1234567890abcdef0  
--description "This is my root volume snapshot"
```

Copy a snapshot and proof of copy

- Use the AWS CLI command **copy-snapshot**.
- Data in transit is automatically encrypted.
- New snapshot has a different ID from the original.

CLI command: Copy a snapshot

```
$ aws ec2 copy-snapshot --region us-east-1 \
--source-region us-west-2 \
--source-snapshot-id snap-1234567890abcdef0 \
--description "This is my copied snapshot"
```

CLI command: Proof of copy to a new Region

```
$ aws ec2 describe-snapshots \
--snapshot-ids snap-0b3c2a7c2a7e4eec6 \
--region us-east-1
```

Restore a snapshot

- Snapshot is stored in Amazon S3 but is not directly accessible. Managed by AWS.
- Find the snapshot ID and restore to a new volume using the **create-volume** command.
- Volumes restored from snapshots have a first-access penalty. Consider reading all the blocks to eliminate penalty in production.

CLI command: Copy a snapshot

```
$ aws ec2 create-volume \  
--size 80 \  
--availability-zone us-east-1a \  
--volume-type gp2 \  
--snapshot-id snap-1234567890abcdef0
```


Restore a snapshot

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CLI command: Proof of copy to a new Region

```
$ aws ec2 \  
describe-volume-status \  
--volume-ids vol-0c108c43c627cb26d
```

Use **Amazon Data Lifecycle Manager** (Amazon DLM) to automate the creation, retention, and deletion of snapshots.

Amazon DLM uses tags to identify EBS volumes to back up.

Define the desired backup and retention actions to Amazon DLM in a lifecycle policy.

Resource type

Target tag

Schedule

EC2 snapshot management using Amazon DLM

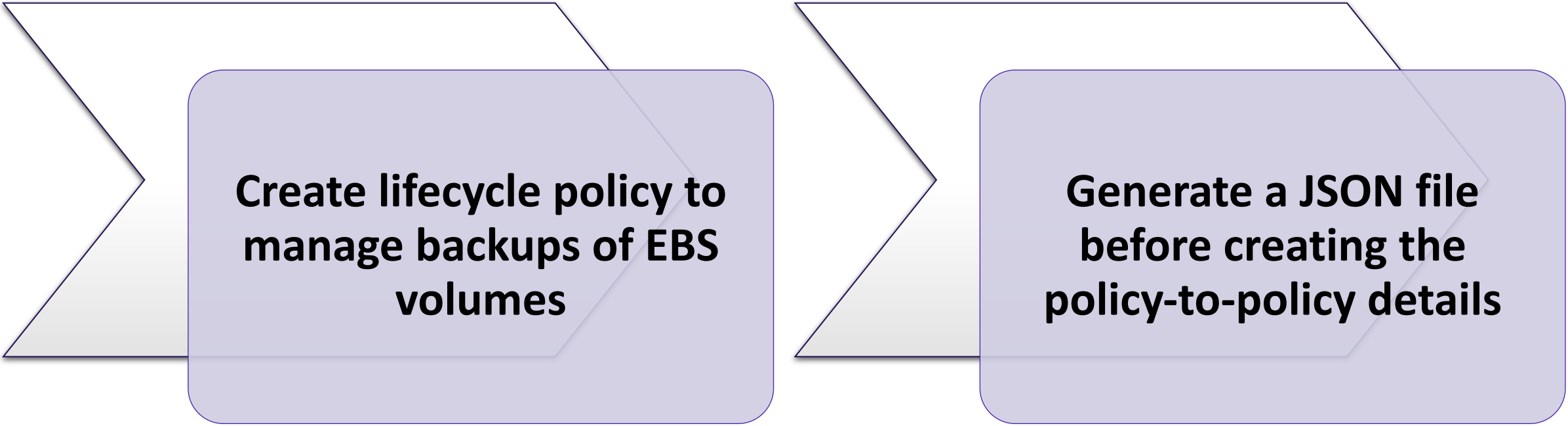


AWS CLI command for EC2 snapshot management using Amazon DLM:

```
$ aws dlm create-default-role
```

Manage EBS volumes with lifecycle policy

Create a **lifecycle policy** to manage the backups of your EBS volumes. A JSON file will need to be generated to cover the details of the policy.



**Create lifecycle policy to
manage backups of EBS
volumes**

**Generate a JSON file
before creating the
policy-to-policy details**

Manage EBS volumes with lifecycle policy

Command to create the lifecycle policy:

```
$ aws dlm create-lifecycle-policy \  
--description "My backup policy" \  
--state ENABLED \  
--execution-role-arn\  
arn:aws:iam::197757065544:role/AWSDataLifecycleManagerDefaultRole \  
--policy-details file://policyDetails.json
```

Key Takeaways



- Amazon EBS is network-attached block storage.
- Amazon EBS is persistent local storage that can store data for applications and databases.
- Amazon EBS snapshots:
 - Incremental
 - Stored in Amazon S3

Module 8: Storage and Archiving

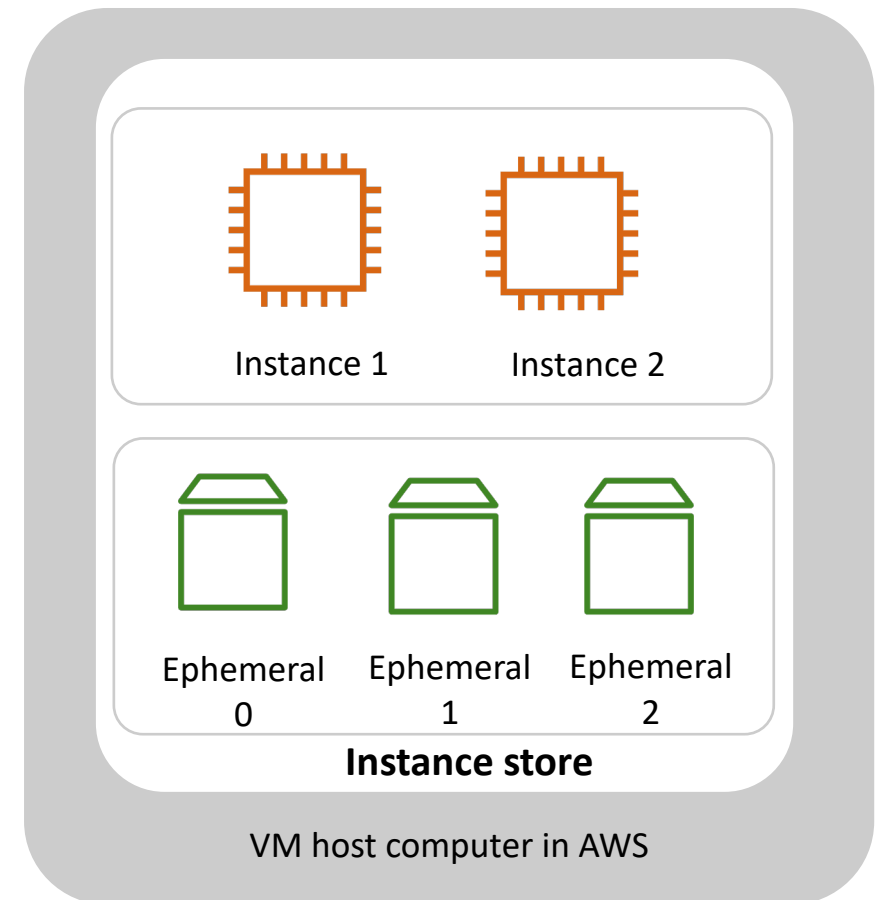
Instance store

Instance store definition:

- **Non-persistent** block-level storage on a shared-disk subsystem
- Fast
- Reclaimed when instance terminated
- Complimentary with certain instances

Use cases:

- Buffers
- Cache
- Scratch data



- Instance store is available for many instance types but not all.
- Number, size, and type (HDD vs. SSD) differs by instance type:
 - Instance stores that use NVMe have higher I/O performance.
- Mounting:
 - Instance store must be mounted before you can access it.
 - Auto-mounted by Windows `EC2Launch` service
 - Typically manually mounted on Linux

Key Takeaways



- **Instance store** is **temporary** block storage physically attached to an instance that is physically attached to a host.
- Use an instance store to store information that does not need to be kept beyond instance life.

Module 8: Storage and Archiving

Amazon Elastic File System



Amazon Elastic File
System

File system storage for Linux-based workloads

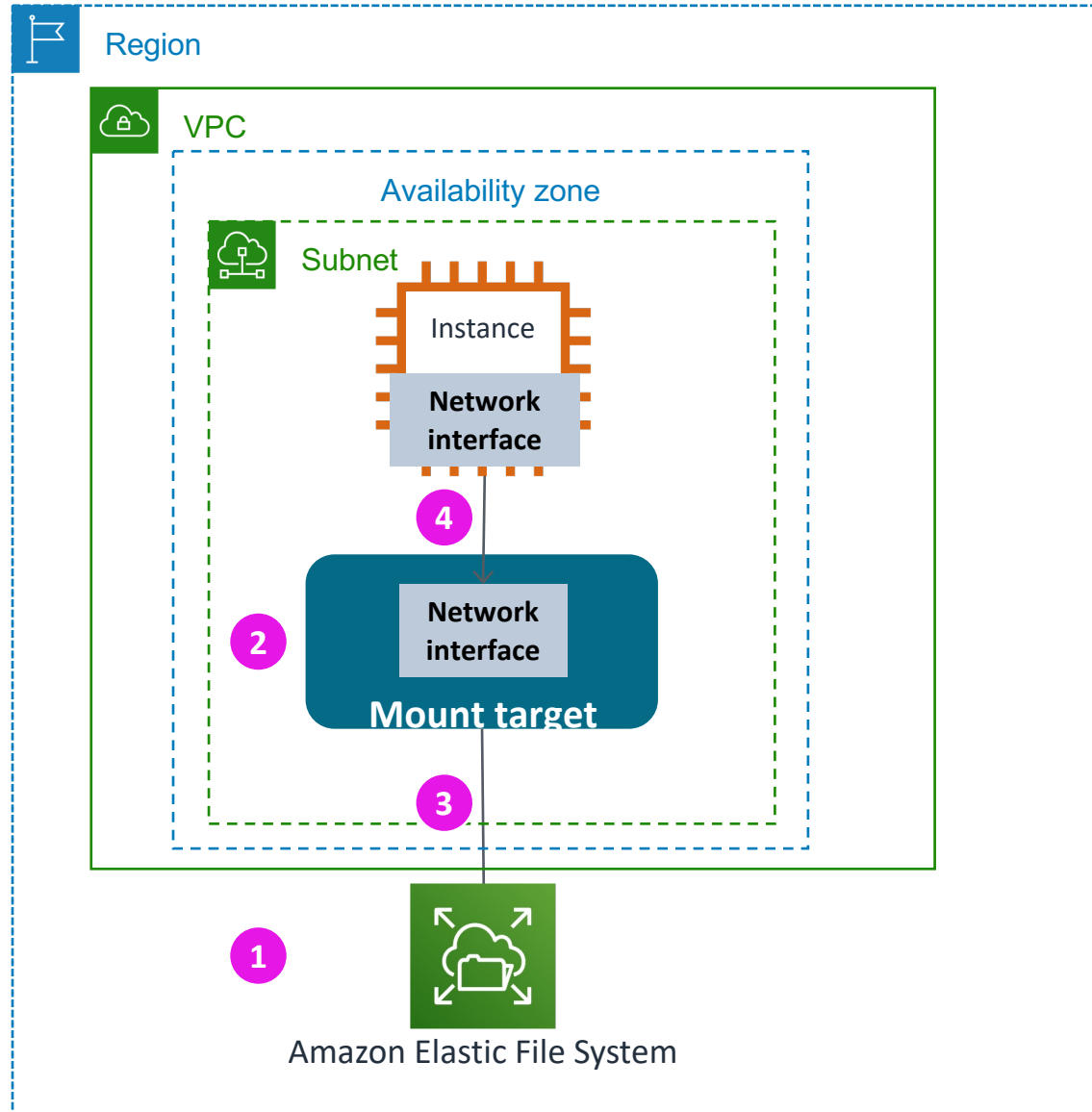
**Shared file
storage**

**Petabyte-scale
file system**

**Supports
Network File
System version
4.0 and 4.1
protocols**

**Compatible with
all Linux-based
AMIs for Amazon
EC2**

Amazon EFS file system setup



Amazon EFS use cases

Amazon EFS is designed to provide performance for a broad spectrum of workloads and applications.

Home directories

File system for enterprise applications

Application testing and development

Database backups

Web serving and content management

Media workflows

Big Data analytics

Amazon EFS performance attributes

Performance modes

- General purpose
- Max I/O

Storage classes

- Standard
- Infrequent access

Throughput modes

- Bursting throughput
- Provisioned throughput

Key Takeaways

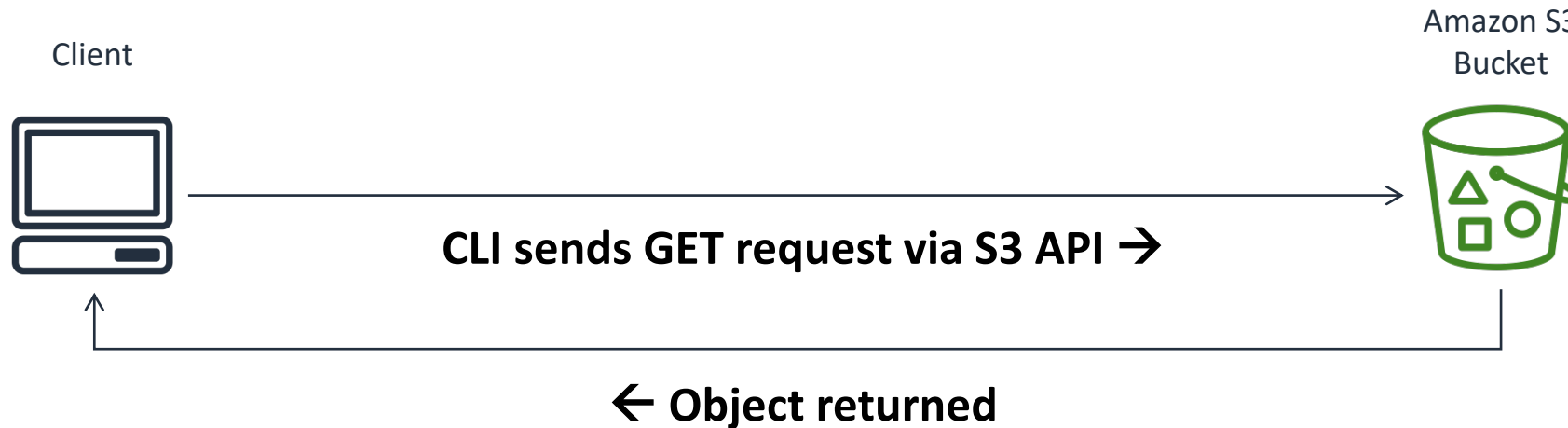


- Amazon EFS is a fully managed file system storage service that uses the NFS protocol.
- Use Amazon EFS to support Linux-based workloads in AWS or on-premises that require shared file system storage.
- The performance of an Amazon EFS file system can be controlled through its performance attributes.

Module 8: Storage and Archiving

Amazon S3

- Fast, durable, highly available key-based access to objects
- Object storage built to store and retrieve data
- Not a file system



Amazon S3 characteristics include:

- Can be accessed anywhere
(both inside and outside of AWS)
- Highly durable (designed for 99.999999999% durability) through redundant storage
- Highly available (designed for 99.99% availability)
- No limit to the amount of data stored
- Up to 5 TB per object



Amazon Simple Storage
Service (S3)

Amazon S3 versioning

Amazon S3 versioning protects objects from accidental overwrites and deletes.

Action	Versioning Enabled	Versioning Disabled/ Versioning Suspended
Upload object with the same Key	Creates a new object with a different version ID. Both are retrievable by version ID.	Overwrites the original object, and previous object is no longer retrievable.
Delete	Adds a Delete marker, but the object is still retrievable by version ID.	Deletes the object, and it is no longer retrievable.

Object storage classes

General purpose

**Amazon S3
Standard**

Intelligent tiering

**Amazon S3
Intelligent-Tiering**

Infrequent access

**Amazon S3
Standard-
Infrequent Access
(S3 Standard-IA)**

**Amazon S3 One
Zone-Infrequent
Access (S3 One
Zone-IA)**

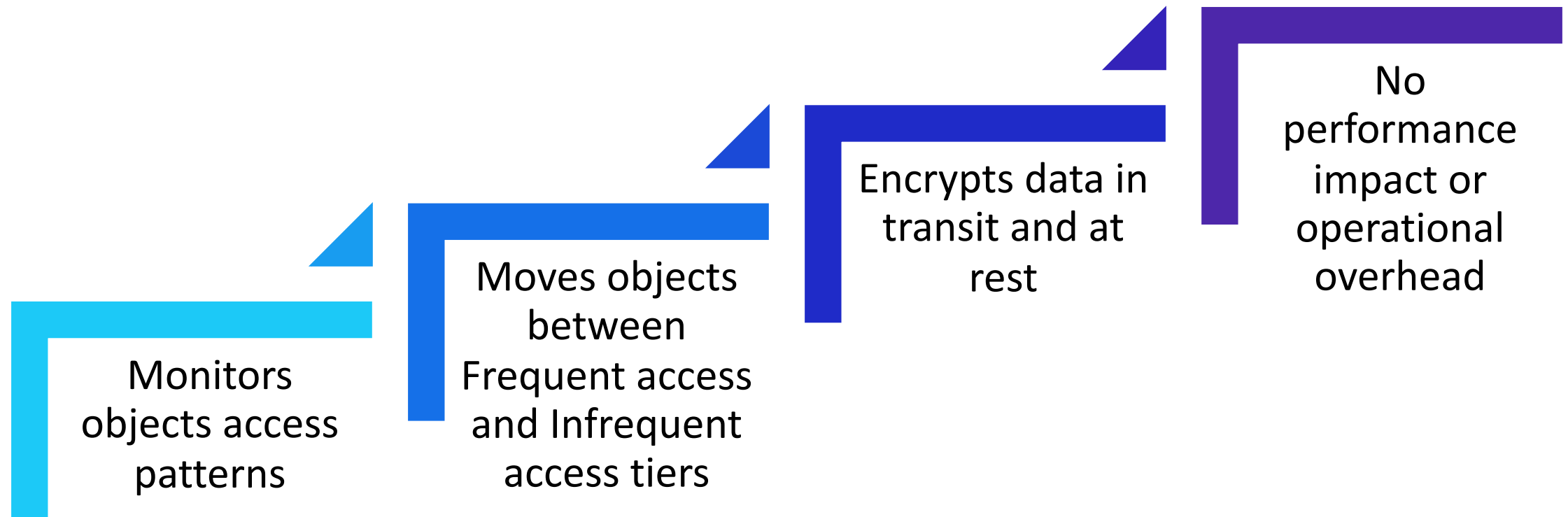
Archive

Amazon S3 Glacier

**Amazon S3 –
Glacier Deep
Archive**

Intelligent tiering

Intelligent tiering automatically moves data to the most cost-effective access tier.



Amazon S3 block public access

Overview

Properties

Permissions

Management

Public access settings

Access Control List

Bucket Policy

CORS configuration

Public access settings for this bucket

Use the Amazon S3 block public access settings to enforce that buckets don't allow public access to data. You can also configure the Amazon S3 block public access settings at the account level. [Learn more](#)

Manage public access control lists (ACLs)	Manage public bucket policies	Edit
Block new public ACLs and uploading public objects <i>(Recommended)</i> True		
Remove public access granted through public ACLs <i>(Recommended)</i> True		
	Block new public bucket policies <i>(Recommended)</i> True	
	Block public and cross-account access if bucket has public policies <i>(Recommended)</i> True	

Amazon S3 public access settings

Block new public ACLs and uploading public objects

Remove public access granted through public ACLs

Block new public bucket policies

Block public and cross-account access to buckets that have public policies

Amazon S3 object lock

- Store objects using “write once, read many” (WORM) model
- Enable at time of bucket creation

**Two ways to
manage object
retention:**

- Retention
- Legal hold

**Two retention
modes:**

- Compliance
- Governance

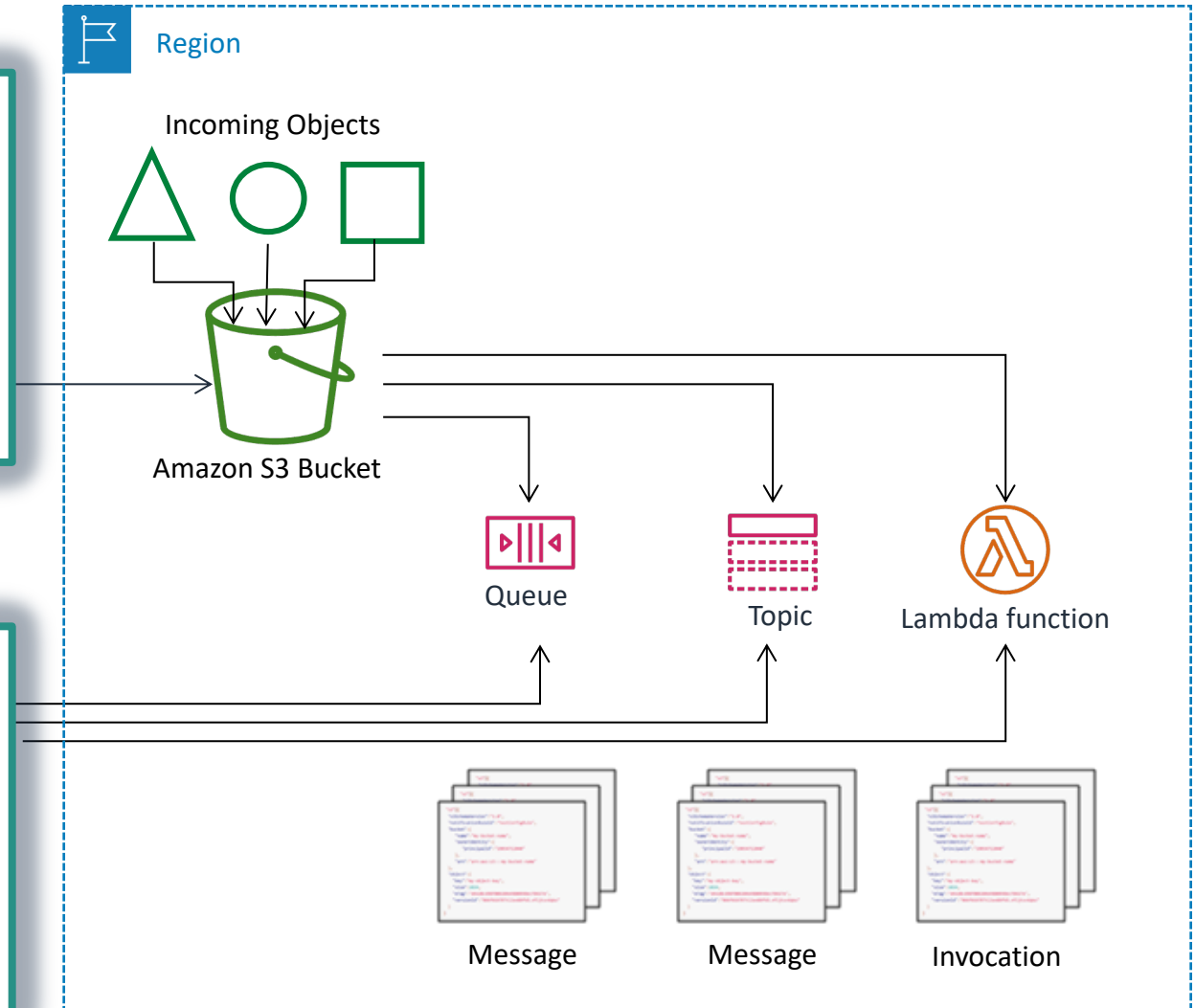
Event notification for Amazon S3

Configure buckets so notifications are issued when an object is:

- Added
- Deleted
- Overwritten

Notifications can be issued directly to:

- Amazon SQS queues
- Amazon SNS topics
- AWS Lambda functions



Additional Amazon S3 features

Feature	Purpose/Benefit	How to use it
Object lifecycle management	To manage your objects so that they are stored cost effectively throughout their lifecycle	Create a <i>lifecycle configuration</i> with rules that define when objects should transition to another storage class and when they should be deleted .
Presigned object URL	To share a private object with someone who does not have AWS security credentials or permissions	Generate a <i>presigned object URL</i> programmatically and provide it to the recipient to access the object.
Cross-Origin Resource Sharing (CORS)	To allow an Amazon S3 bucket hosting a static website to support CORS Many origins to one bucket	Create a CORS configuration on the bucket with rules that identify the authorized origins and HTTP operations.

Create and list Amazon S3 buckets

- Create a new bucket:

```
aws s3 mb s3://mybucket
```

- List all buckets in the AWS account:

```
aws s3 ls
```

Work with Amazon S3 objects

- Copy a local file to Amazon S3:

```
$ aws s3 cp file.txt s3://mybucket/myprefix/file.txt
```

- Sync objects under a specified bucket and prefix to the current local directory:

```
$ aws s3 sync s3://mybucket/myprefix/
```

- Remove an object from a bucket:

```
$ aws s3 rm s3://mybucket/myprefix/file.txt
```

Create an object with a specific storage class

Use the **--storage-class** parameter with the copy command to specify the storage class.

```
$ aws s3 cp file.txt s3://mybucket/myprefix/file.txt \  
--storage-class INTELLIGENT_TIERING
```

The s3api command:

- Provides direct access to the Amazon S3 APIs
- Enables operations not exposed by **aws s3**

```
aws s3api
```

```
create-multipart-upload
```

```
put-object-acl
```

```
put-bucket-policy
```

```
list-object-versions
```

Key Takeaways



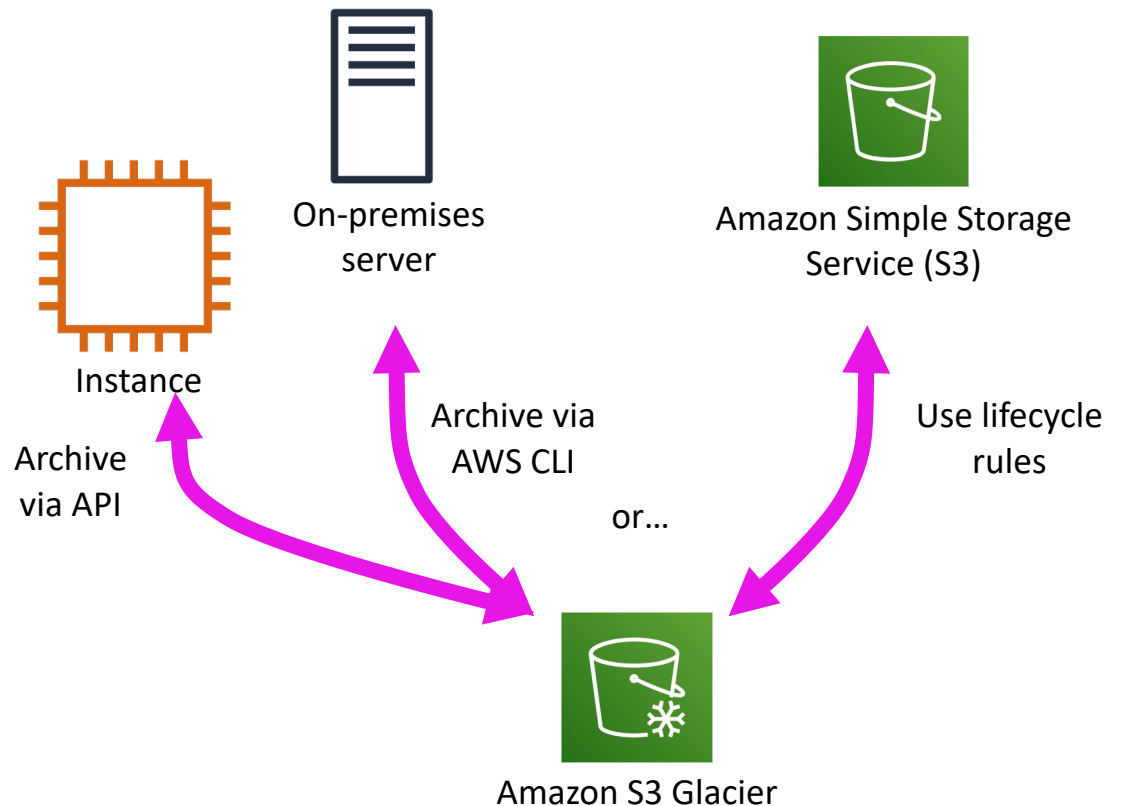
- Amazon S3 is a regionally provisioned object storage service.
- Use Amazon S3 for durability and high availability.
- Amazon S3 offers multiple storage classes, versioning, object locking, and event notification features.

Module 8: Storage and Archiving

Amazon S3 Glacier

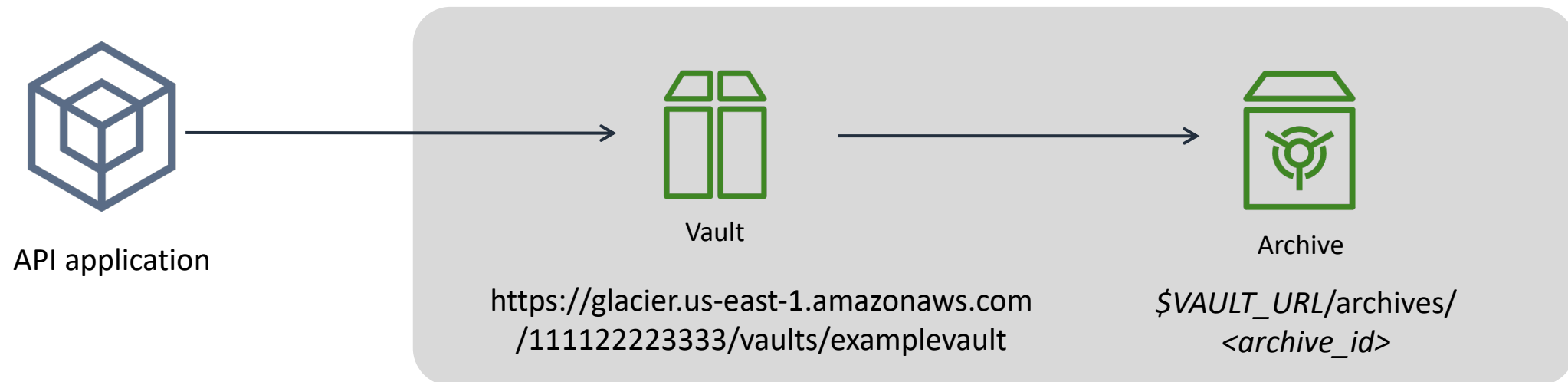
Amazon S3 Glacier

- Extremely low-cost data archiving and long-term backup
- Retrieval by request
- Can configure lifecycle archiving of Amazon S3 content to Amazon S3 Glacier



How Amazon S3 Glacier stores data

- Amazon S3 Glacier stores data in **an archive**.
 - A basic unit of data in Amazon Glacier (document, video, image, etc.)
- A **vault** is a collection of archives.



Interact with Amazon S3 Glacier using:

Amazon S3 lifecycle policies

- Archived files are accessed via the Amazon S3 console or Amazon S3 API

Amazon S3 Glacier API

- Directly add files
- To retrieve files:
 1. Initiate a job request.
 2. Select the archive retrieval operation.
 3. Optionally specify a date range or a byte range filter.
 4. Poll for job completion, or receive SNS notification.

Archive retrieval options

Expedited
1-5 minutes

Standard
3-5 hours

Bulk
5-12 hours

Key Takeaways



- Amazon S3 Glacier is a long-term storage service for content that does not need to be immediately available.
- Use Amazon S3 Glacier as an affordable long-term storage solution for data archiving or backup.
- Amazon S3 Glacier offers three archive retrieval options.

Module 8: Storage and Archiving

AWS Storage Gateway



AWS Storage Gateway

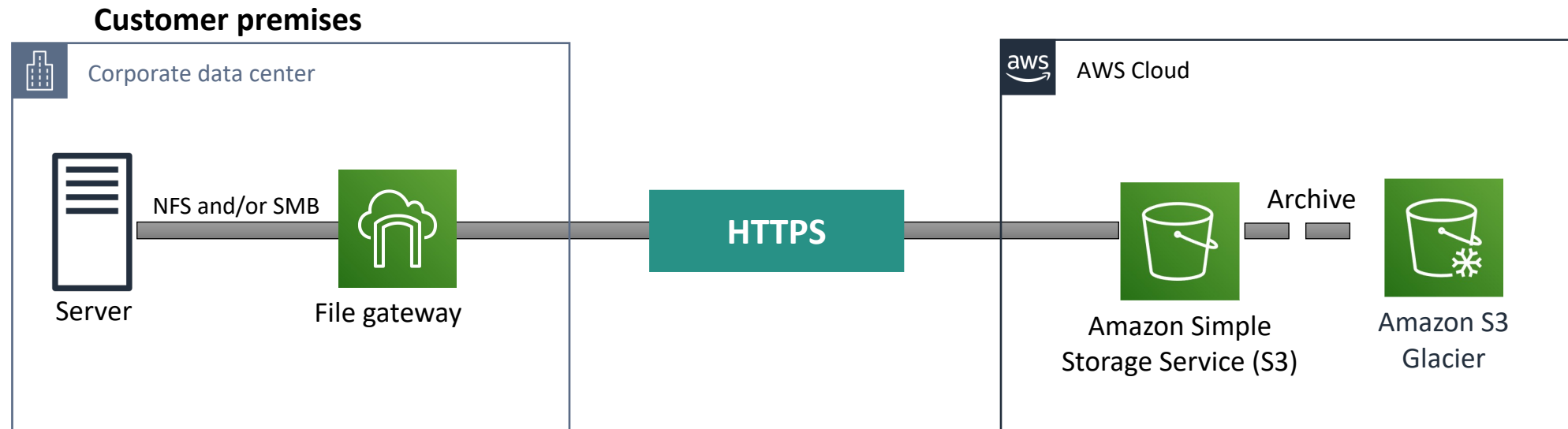
Hybrid cloud storage solution that allows on-premises applications to access data in AWS Cloud.

File gateway

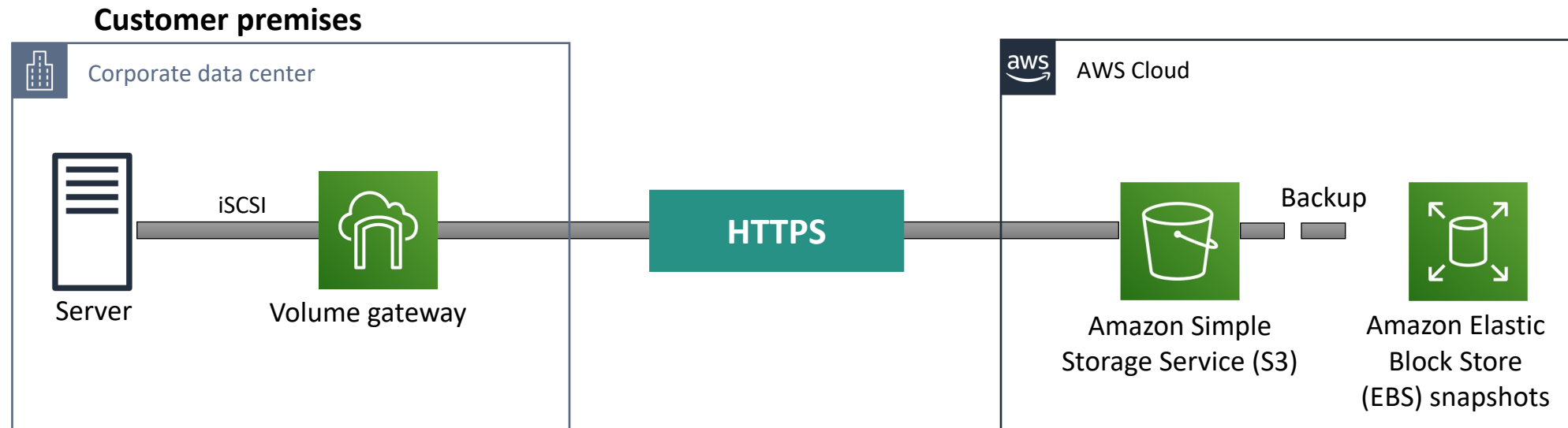
Volume gateway

Tape gateway

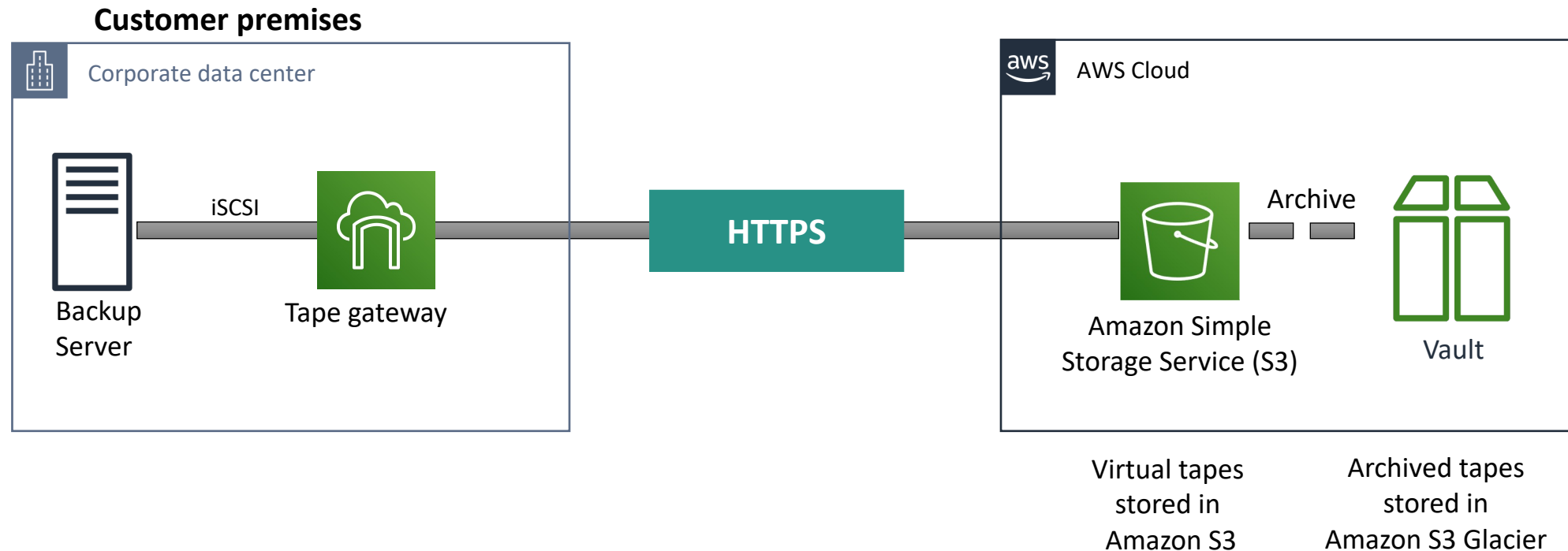
File gateway



Volume gateway



Tape gateway



Key Takeaways



- AWS Storage Gateway sits on-premises and links the environment to the AWS Cloud.
- Use AWS Storage Gateway for hybrid scenarios where some storage is needed on-premises. Offload to Amazon S3, Amazon S3 Glacier, or Amazon EBS.
- AWS Storage Gateway provides three options—file, volume, and tape.

Module 8: Storage and Archiving

AWS data transfer and migration services

- Managed, highly available SFTP server used to send data to and from Amazon S3



AWS Transfer
for SFTP

Secure
transfer using
SFTP into and
out of AWS

Retain
existing
workflows

Simple to use

Data stored in
Amazon S3
bucket



AWS DataSync

Efficient and
fast

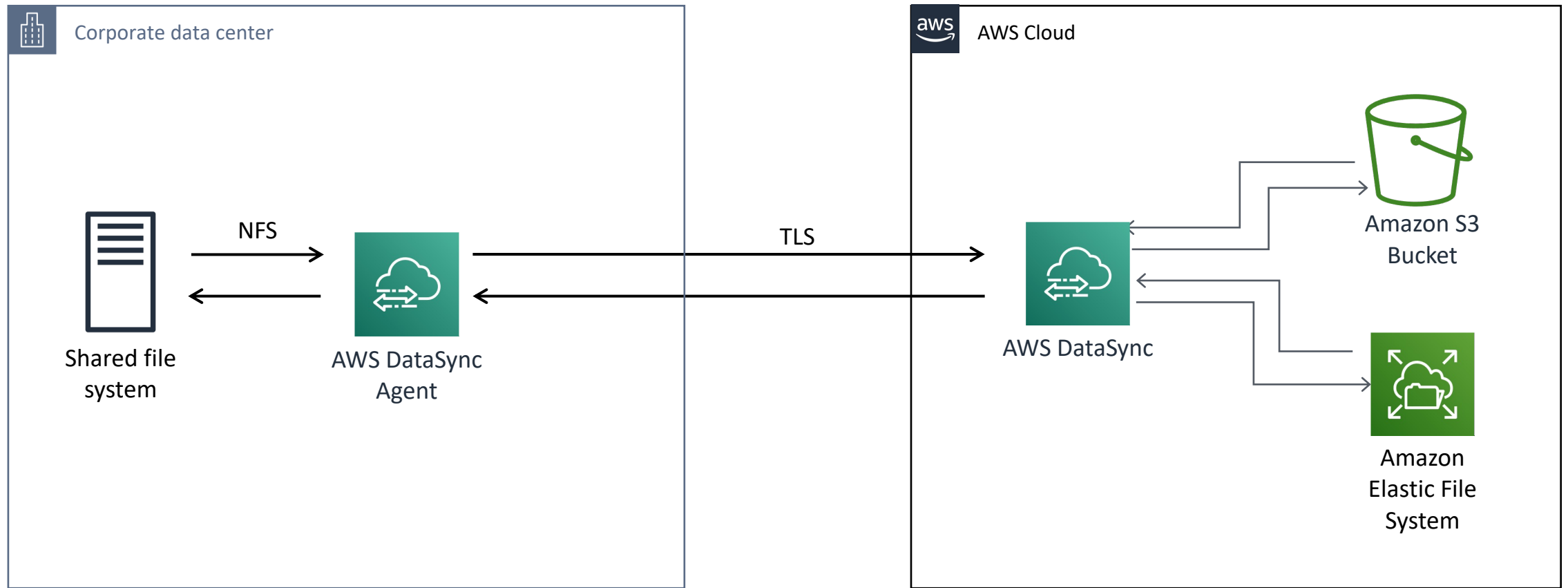
Managed
service

AWS DataSync
Agent (NFS
protocol)

Sync between on-premises and
AWS

Connect via
internet or AWS
Direct Connect

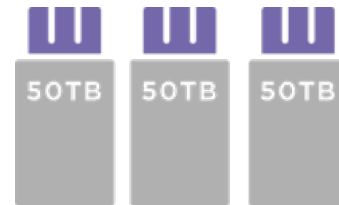
Using AWS DataSync



AWS Snowball



Secure enclosure



50/80TB
Shipped in parallel



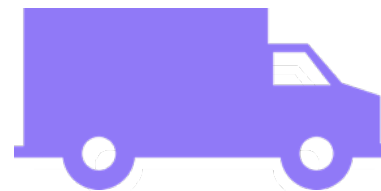
Simplified logistics



Strong encryption,
end-to-end



Large customer
dataset



End-to-end
Custody (escorted)



Customer dataset
loaded



In summary, you learned about:

- AWS data storage options, and their purpose and benefits
- How to create and manage Amazon EBS snapshots
- How to store, retrieve, and archive Amazon S3 objects
- Main AWS data migration services

To finish this module:

Complete:  **Knowledge Check**

Test Strategies



Module 8 Scenario:

The local power company has recently installed a new power grid within the community. They are working with the local government to acquire funding for a second power grid, but must show data that shows the cost reductions and how that translates to increased consumption time windows. This data needs to be kept for a minimum of 10 years in case of audit or by inquiry of a government agency. The power company wants to engage the community and gain their support, as their individual and communal power consumption is reflected in the data.

The CEO has decided that the company will now begin processing the big data in real time through the use of Amazon Kinesis. The architecture has been developed with an intermediate and backup Amazon Simple Storage Service (Amazon S3) bucket for potential delivery failures and source records.

What other services should be used for transformed records after they are processed by Amazon Kinesis Data Firehose? Select two responses.



Module 8 Scenario:

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What other services should be used for transformed records after they are processed by Amazon Kinesis Data Firehose? Select two responses.

- A. Amazon Elastic Block Store (Amazon EBS)
- B. Amazon Elasticsearch Service (Amazon ES)
- C. Amazon Glacier
- D. Amazon Simple Storage Service (Amazon S3)
- E. Amazon Redshift cluster

Lab 5: Managing Storage in AWS (Linux)

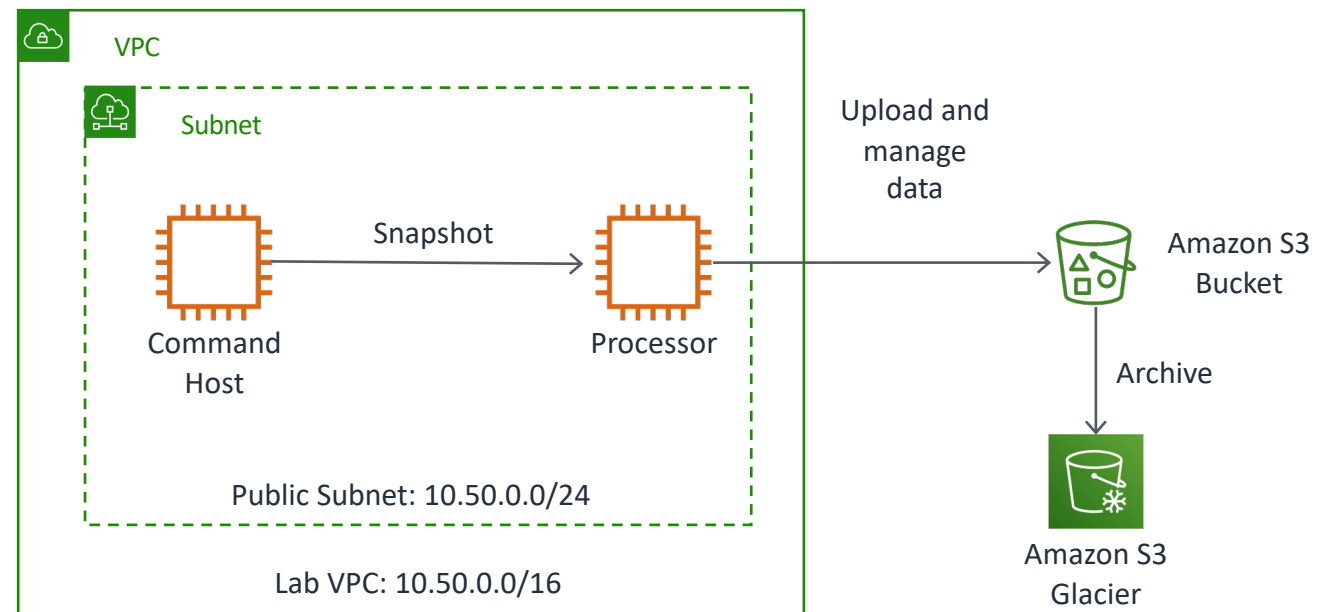


~ 45 minutes

Lab 5: Managing storage in AWS

In this lab, you will create and maintain Amazon EBS snapshots by completing the following tasks:

1. Create an Amazon EC2 instance.
2. Create an Amazon S3 bucket.
3. Create Amazon EBS snapshots.
4. Schedule snapshot creation.

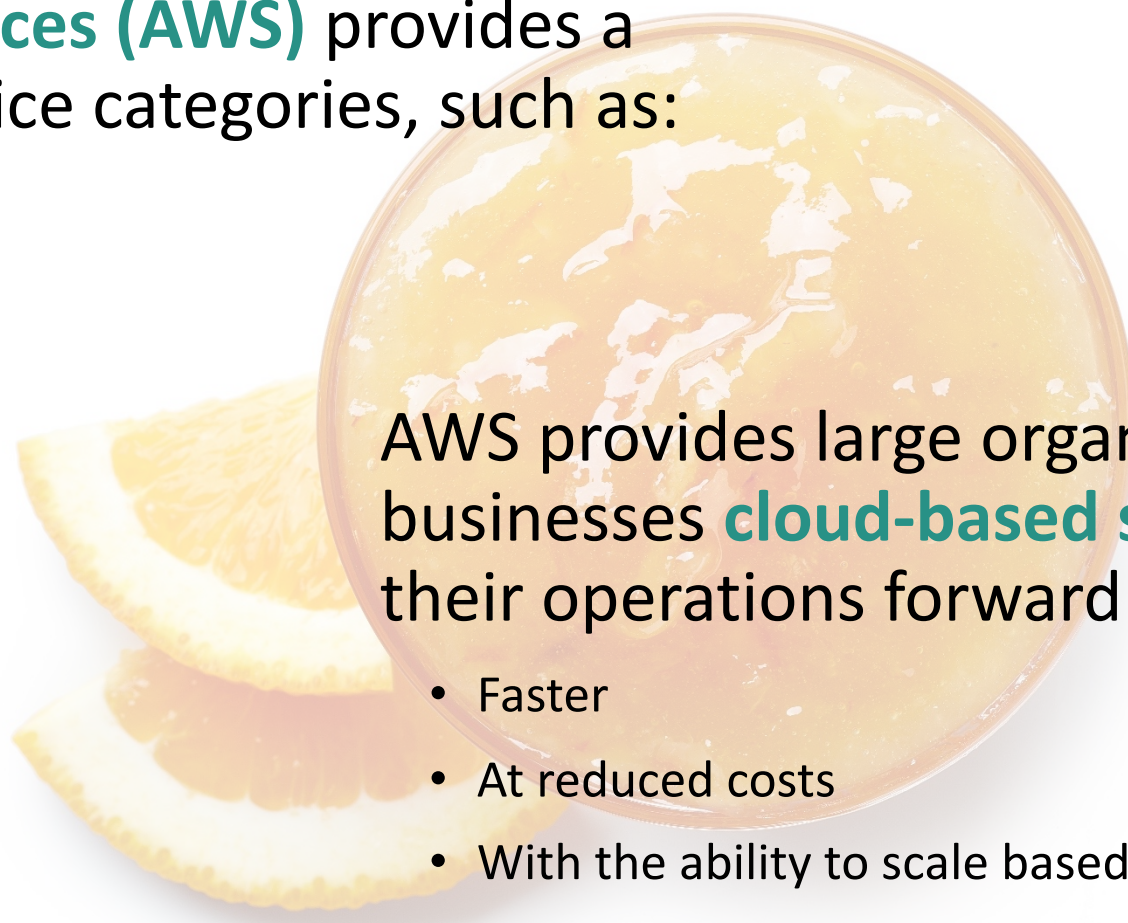


Systems Operations Project Scenario

Mom & Pop Café & Bakery

Amazon Web Services (AWS) provides a broad array of service categories, such as:

- Compute
- Storage
- Database
- Analytics

A decorative image featuring a glass bowl filled with orange juice and several orange slices floating in it, positioned behind the text on the right side of the slide.

AWS provides large organizations and small businesses **cloud-based solutions** to move their operations forward:

- Faster
- At reduced costs
- With the ability to scale based on demand

On completion of this project, you will be able to:

- **Experience** the communication challenges faced when attempting to apply technology as the solution to business problems.
- **Translate** customer requirements into a proposed technical solution.
- **Present** the proposed solution to the customer.
- **Use** AWS Best Practices for a frugal, resilient, and secure solution.



Customer Meeting Preparation

Mom & Pop Café & Bakery

Working with S3

Business need: Sharing files with an external partner

Natalia suggests **using Amazon Simple Storage Service (Amazon S3)** to share files with an external partner.



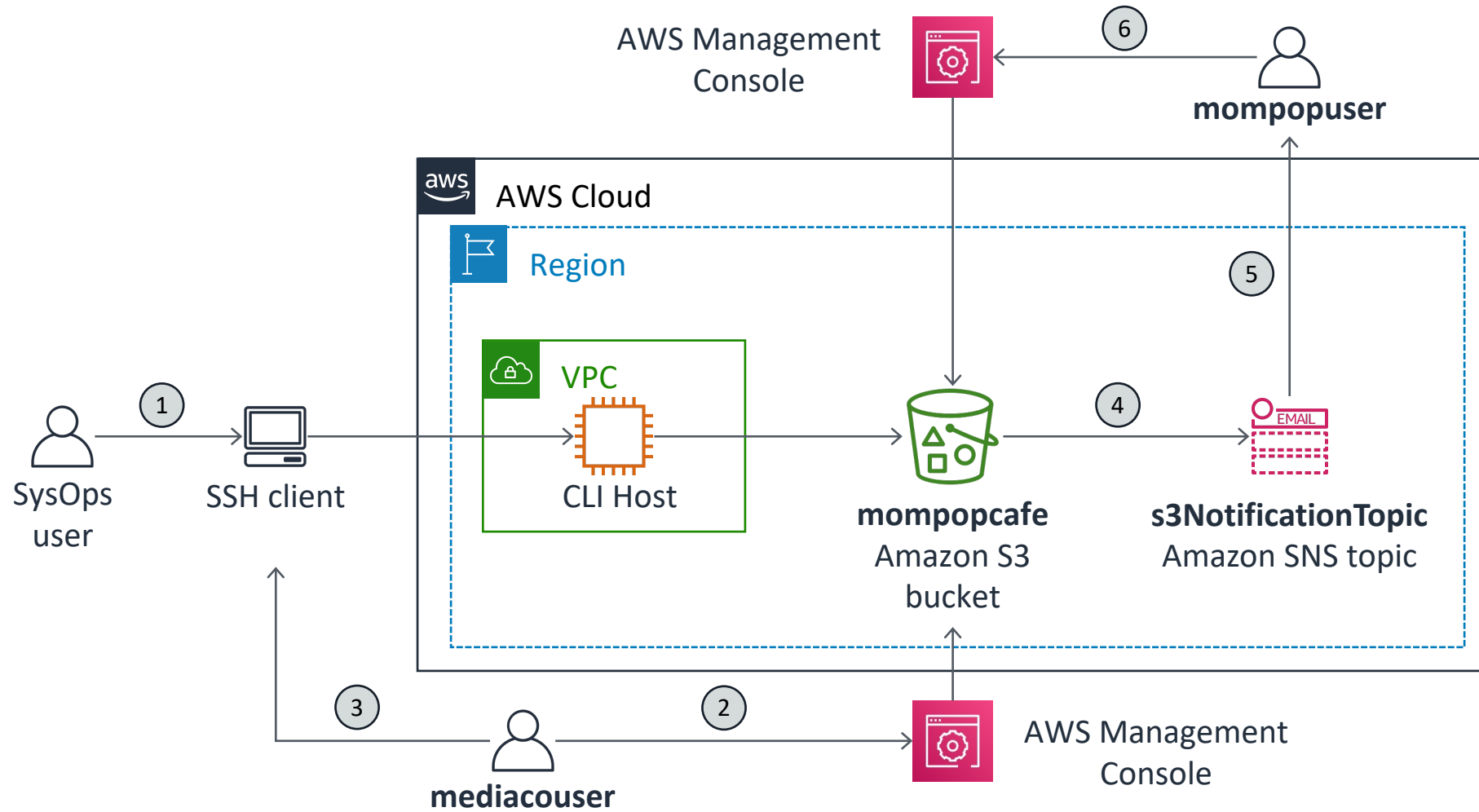
Natalia discusses Amazon S3 file sharing benefits



Natalia discusses the **benefits of Amazon S3** for storing image files:

1. Share access with other users.
2. Configure different levels of access security.
3. Create event notifications.
4. Store a lot of data in a reliable, fast, and redundant way.

Activity 8: Work with Amazon S3





Begin Activity #8: Work with Amazon S3

Benefits of using S3 for sharing files

What are two benefits of using Amazon S3 to share files?

Answer #1

Answer #2

Benefits of using S3 for sharing files

What are two benefits of using Amazon S3 to share files?

Answer #1

Configure different levels of access security.

Answer #2

Create event notifications.

Project time: Troubleshooting knowledge base



Thank You

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